

QUANTITATIVE METHODS II

Practice Problems · Topics 1 – 9

TOPIC 1 STATISTICS, DATA & STATISTICAL THINKING

◆ Q 1A

A hospital administrator surveyed 25 patients about their preferred method of receiving appointment reminders. The results were:

SMS Email Phone Call SMS Email
Email SMS SMS Phone Call Email
Phone Call Email SMS Email Phone Call
SMS Email Phone Call SMS Email Phone Call
Email SMS Phone Call

- a) Identify the type of data (qualitative/quantitative, primary/secondary, raw/aggregate, cross-section/time series).
- b) What is the population and what is the sample in this study?

◆ Q 1B

A sample of 8 e-commerce companies was collected. The table below shows selected variables:

Company	Revenue (\$B)	Rank	Employees	Net Income (\$B)	Industry Code
Amazon	514.0	1	1,540,000	30.4	3
Alibaba	126.5	2	235,000	10.2	3
Shopify	7.1	3	12,000	1.3	4
eBay	9.8	4	11,600	2.5	3
Etsy	2.6	5	2,800	0.6	4
Wayfair	12.2	6	16,000	-0.2	5
Rakuten	14.1	7	28,000	0.4	3
Zalando	10.4	8	17,000	-0.1	5

- a) How many elements and variables are in the data set?
- b) Classify each variable as qualitative or quantitative.

- c) Which variables are discrete and which are continuous?

◆Q 1C — Descriptive vs. Inferential

Classify each statement as descriptive or inferential statistics and explain:

- a) Based on 400 surveyed students, researchers estimate that 68% of university students sleep fewer than 7 hours per night.
- b) The average monthly electricity bill for all apartments in a building is \$142.
- c) A sample of 150 transactions shows that the average purchase value at a retail store is \$87.
- d) Data from 800 employees reveals that 55% prefer hybrid work arrangements.

◆Q 1D — Sampling Bias

For each scenario, identify whether it involves selection bias, nonresponse bias, or measurement error:

- a) A nutrition survey asks participants to self-report their daily calorie intake, but many underreport unhealthy foods.
- b) A satisfaction survey is emailed to 3,000 customers; only 310 respond.
- c) A study on exercise habits is conducted by surveying members at a fitness centre.

TOPIC 2 MEASURES OF CENTRAL TENDENCY

◆Q 2A

Ten delivery drivers were randomly selected, and their daily delivery counts were:

34, 41, 38, 41, 47, 36, 41, 52, 38, 44

Compute the:

- Mode
- Mean
- Median

◆Q 2B

The following table shows the annual revenue (in millions of dollars) for the 8 largest fast-food chains. Find the mean annual revenue.

Chain	Annual Revenue (\$M)
McDonald's	23,200

Chain	Annual Revenue (\$M)
Starbucks	18,600
Chick-fil-A	12,400
Taco Bell	11,800
Burger King	10,300
Subway	9,400
Domino's	8,600
KFC	7,900

◆Q 2C — Grouped Mean

The table below shows the distribution of weekly study hours for a sample of university students. Estimate the mean study hours.

Study Hours	Midpoint	Frequency
0 – 4	2	45
5 – 9	7	120
10 – 14	12	185
15 – 19	17	160
20 – 24	22	95
25 – 29	27	55
30 – 34	32	30

◆Q 2D — Stem-and-Leaf & Histograms

The following data shows weekly training hours for 15 call centre agents:

2, 4, 7, 4, 9, 11, 7, 6, 4, 8, 10, 7, 5, 4, 9

- Find the mean, median, and mode
- Construct a stem-and-leaf plot
- Create a frequency table
- Draw a histogram (frequency, relative frequency, and cumulative frequency)

◆Q 2E — Five-Number Summary

Daily orders processed by a warehouse (hundreds) for 15 randomly selected days:

3.8, 5.5, 7.2, 5.5, 8.9, 10.4, 7.8, 6.6, 5.5, 9.1, 9.8, 8.9, 7.2, 5.5, 9.5

- Find Q1, Q2 (median), and Q3
- Find the range and IQR
- Write the five-number summary
- Draw a box plot

TOPIC 3 MEASURES OF VARIABILITY

◆Q 3A

Eight interns at a consulting firm were randomly selected, and their monthly stipends (in \$000s) were:

6, 9, 12, 9, 15, 8, 11, 10

Compute the:

- Sample Variance
- Sample Standard Deviation (using both formulas)

◆Q 3B — Consistency Comparison

Two coffee shops track wait time (in seconds) for 5 randomly sampled customers:

Shop A: 88, 89, 90, 91, 92

Shop B: 70, 82, 90, 98, 110

Both shops have a mean wait time of 90 seconds.

- Calculate the range, variance, and standard deviation for each shop.
- Which shop provides more consistent service? Explain.

TOPIC 4 PROBABILITY

◆Q 4A — Basic Probability

A bag of assorted fruit sweets was opened and the pieces counted:

Flavour	Count
Raspberry	35
Mango	28
Watermelon	22
Peach	18

Flavour	Count
Blueberry	12

- List the sample space for this problem.
- Find the probability of randomly selecting a blueberry sweet.
- Find the probability of selecting a raspberry or mango sweet.
- Find the probability of NOT selecting a watermelon sweet.
- Find the probability of selecting either peach or blueberry sweet.

◆Q 4B — Union & Intersection

A number is randomly chosen from 1 to 15, inclusive.

A: The number is a multiple of 4

B: The number is greater than 9

- Identify all sample points in $A \cup B$.
- Identify all sample points in $A \cap B$.
- Find $P(A \cup B)$ and $P(A \cap B)$.

◆Q 4C — Contingency Table

A survey of 500 employees examined their contract type and remote work frequency:

Contract / Remote Work	Never	Sometimes	Always	Total
Full-time	150	80	20	250
Part-time	60	70	20	150
Freelance	20	50	30	100
Total	230	200	70	500

- What is the probability the employee is a freelancer AND always works remotely?
- What is the probability the employee is full-time OR never works remotely?
- Given that the employee sometimes works remotely, what is the probability they are part-time?

◆Q 4D — Independent Events

A start-up launches two products in successive quarters. $P(\text{Product A succeeds}) = 0.65$ and $P(\text{Product B succeeds}) = 0.55$. Assume the events are independent.

- What is the probability both products succeed?
- What is the probability at least one product succeeds?

- c) Can these two events be mutually exclusive? Explain using the Additive Rule.

TOPIC 5 PERMUTATIONS & COMBINATIONS

◆Q 5A

1. In how many ways can 6 runners be arranged in a starting line?
2. A manager randomly selects 4 employees from a team of 10 to attend a conference. How many different groups are possible?
3. A digital safe uses a 5-digit code (digits 0–9). How many possible codes exist?
4. A set menu offers 3 soups, 5 mains, and 4 desserts. How many different meals can be ordered?
5. In how many ways can 6 different trophies be arranged on a shelf?
6. A working committee of 3 must be selected from 8 candidates. How many different committees are possible?
7. In how many ways can the letters of the word MARKET be arranged?
8. A club must elect a chairperson, a secretary, and a treasurer from 15 members. In how many ways?

◆Q 5B — Passwords & Mixed Arrangements

You are setting a password using digits (0–9) and/or letters (a–z). Find the number of possible passwords for each case:

- Any 5 digits (repetition allowed)
- Any 4 letters (repetition allowed)
- Any 3 letters followed by 3 digits (repetition allowed)
- 5 different digits (no repetition)
- 4 different letters (no repetition)
- 2 letters followed by 3 digits (no repetition)

In how many ways can the letters in the word MANAGEMENT be arranged?

A committee of 5 people is to be formed from 7 men and 6 women. In how many ways can they be selected such that at least 3 women are included?

TOPIC 6 DISCRETE RANDOM VARIABLES & PROBABILITY DISTRIBUTIONS

◆Q 6A — Validity & Basic Calculations

Determine whether the following table represents a valid probability distribution. Explain your answer.

x	P(x)
0	0.10
1	0.30
2	0.35
3	0.20
4	0.10

If valid, find:

- $P(X = 3)$
- $P(X \geq 2)$
- $P(X < 4)$

◆Q 6B — Expected Value & Variance

A ride-sharing company tracks the number of trips completed per hour by a driver:

x (trips)	P(x)
0	0.08
1	0.20
2	0.38
3	0.22
4	0.12

- a) Compute $E(X)$ — the expected number of trips per hour.
- b) Compute $\text{Var}(X)$ and the standard deviation.
- c) If the driver earns \$12 per trip, what is the expected hourly earnings?

◆Q 6C — Game & Decision Making

A raffle game has the following outcomes:

Outcome	Probability	Net Payoff (\$)
Grand Prize	0.02	+200
Minor Prize	0.18	+15
No Prize	0.80	-5

- a) Find the expected payoff per game.
- b) Is this game fair? Explain.
- c) How much would the ticket price need to be to make the game exactly fair?

TOPIC 7 BINOMIAL DISTRIBUTION

◆Q 7A — Basic Binomial Problems

1. A fair die is rolled 8 times. What is the probability of getting exactly 3 sixes?
2. A student guesses randomly on a true/false quiz with 7 questions. What is the probability of getting exactly 5 correct?
3. A sales rep closes a deal with probability 0.35 on any given meeting. In 10 meetings, what is the probability of closing exactly 4 deals?
4. A supplier delivers defective items with probability 0.10. Out of 8 sampled, what is the probability that at most 1 is defective?
5. $P(\text{5-star review}) = 0.60$. If 12 customers respond independently, what is the probability that at least 9 leave 5-star reviews?
6. A student has $P(\text{on-time submission}) = 0.80$. Over 10 assignments, what is the probability of exactly 7 on-time submissions?

◆Q 7B — Applied Binomial

A health agency reports that 10% of people in a region test positive for a seasonal virus. A clinic tests 30 patients in one day.

- a) What is the probability that exactly 4 patients test positive?
- b) What is the probability that at most 2 patients test positive?
- c) What is the probability that at least 5 patients test positive?
- d) What is the expected number of positive tests?
- e) Using the normal approximation, form an interval (± 2 standard deviations) likely to contain the count of positive tests.

TOPIC 8 CONTINUOUS RANDOM VARIABLES

◆Q 8A — Uniform Distribution

1. $x \sim \text{Uniform}(c = 10, d = 50)$. Find $f(x)$, the mean, and standard deviation of x .
2. $x \sim \text{Uniform}(c = 8, d = 32)$. Find $P(14 < x < 26)$.
3. $x \sim \text{Uniform}(c = 30, d = 70)$. Find $P(x > 55)$.

◆Q 8B — Standard Normal Distribution

1. Find $P(-1.20 < Z < 1.85)$.
2. Find z_0 such that $P(-1.0 < Z < z_0) = 0.8500$.
3. Find $P(Z > 1.64)$.
4. Find z_0 such that $P(Z < z_0) = 0.0250$.

◆Q 8C — Normal Distribution Applications

1. $X \sim N(\mu = 72, \sigma = 10)$. Find $P(60 < X < 85)$.
 2. $X \sim N(\mu = 100, \sigma^2 = 36)$. Find x_0 such that $P(X \geq x_0) = 0.0668$.
 3. A vending machine dispenses drinks with mean 355 ml and SD 6 ml (Normal model).
 - In what interval would you expect the central 95% of volumes to fall?
 - What percent of drinks contain more than 367 ml?
 - What percent contain between 343 ml and 361 ml?
 - Below what volume would the worst 2.5% of pours fall?
-

◆Q 8D — Normal Approximation to Binomial

A company reports that 15% of customer service calls result in escalation. In a random sample of 200 calls:

- a) Use the Normal approximation to find $P(\text{at least 35 calls are escalated})$.
- b) Should management be concerned if 45 calls are escalated? Justify using probability.

TOPIC 9 SAMPLING DISTRIBUTION

◆Q 9A — Sampling Distribution Setup

Population distribution:

x	2	4	6
p(x)	1/3	1/3	1/3

Random samples of $n = 2$ are taken from this population.

- a) List all possible samples and their means.
 - b) Find the sampling distribution of $\bar{x}$.
 - c) Construct a probability histogram for $\bar{x}$.
 - d) Find $E(\bar{x})$ and verify that $\bar{x}$ is an unbiased estimator of μ .
 - e) Find the standard deviation of the sampling distribution of $\bar{x}$.
-

◆Q 9B — Central Limit Theorem

Daily customer complaints at a telecom company: skewed distribution with $\mu = 120$, $\sigma = 25$. A sample of 49 days is selected.

- a) Describe the sampling distribution of $\bar{x}$ (state μ_x and σ_x).
 - b) What is the probability that the sample mean exceeds 126 complaints?
 - c) What is the probability that the sample mean falls between 115 and 125 complaints?
-

◆Q 9C — Applied Sampling Problem

Order processing time: right-skewed, $\mu = 48$ min, $\sigma = 18$ min. A quality auditor selects 36 orders.

- a) What is the standard error of the mean?
 - b) What is the probability that the sample mean is greater than 52 minutes?
 - c) What is the probability that the sample mean is between 44 and 51 minutes?
 - d) The company claims average processing time is under 48 minutes. If $\bar{x} = 54$ minutes, is this consistent with the claim? Justify using probability.
-

◆Q 9D — Comparing Groups

Two call centres compared on average handling time (minutes) per call:

Call Centre	Population Mean (μ)	Population SD (σ)	Sample Size (n)
Centre A	6.2	1.8	36
Centre B	7.1	2.4	64

- a) Describe the sampling distribution of $\bar{x}$ for each centre (state μ_x and σ_x).
- b) What is the probability that Centre A's sample mean is less than 5.8 minutes?
- c) What is the probability that Centre B's sample mean is greater than 6.7 minutes?

— End of Practice Problems —